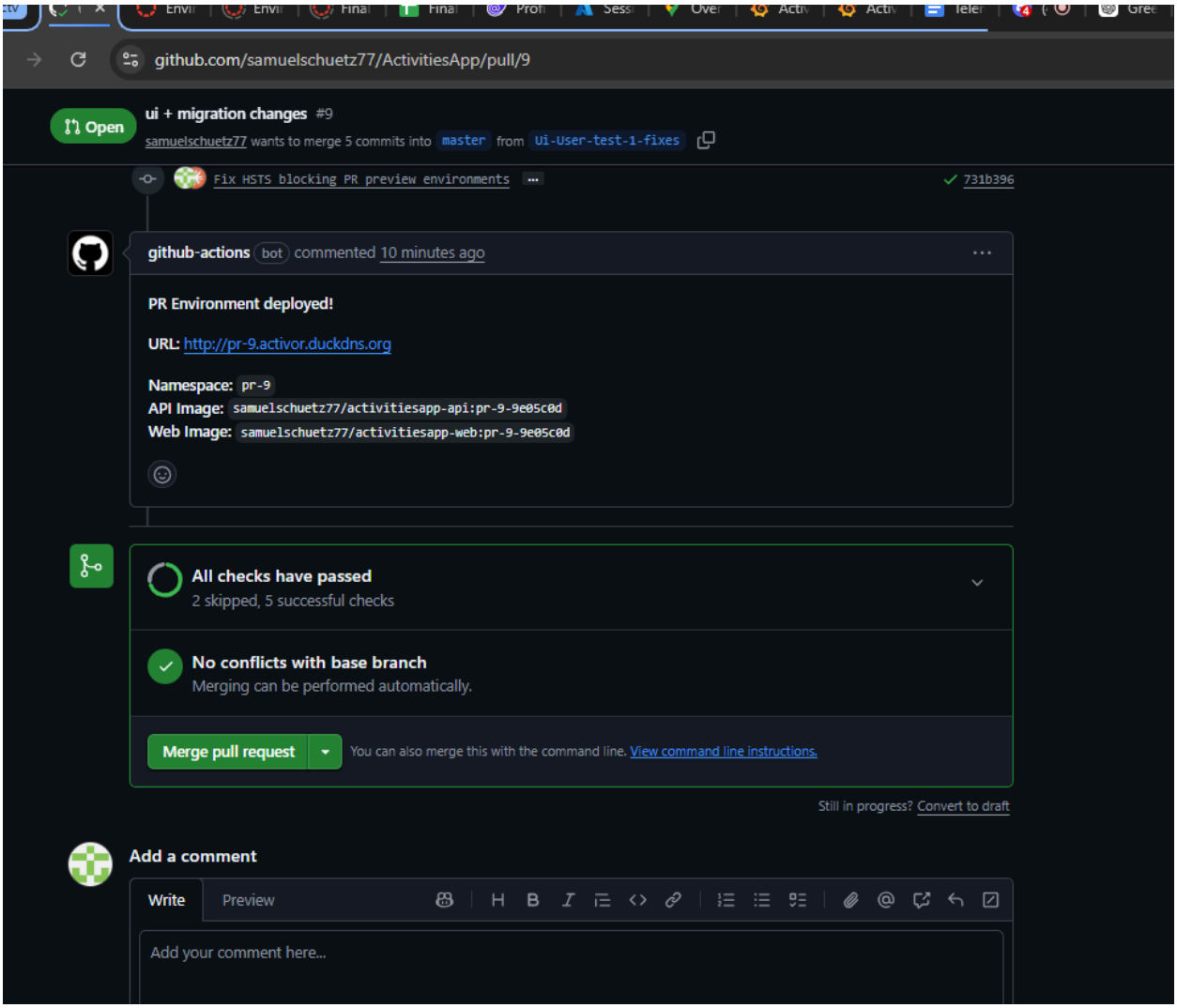
Assignment: Environments for each PR

url posted as comment on github PR page:



deployment pipeline for PR to create new kubernetes environemnt:

```
deploy:
    name: Deploy PR Environment
    if: github.event.action != 'closed'
    needs: push
    runs-on: self-hosted
    env:
```

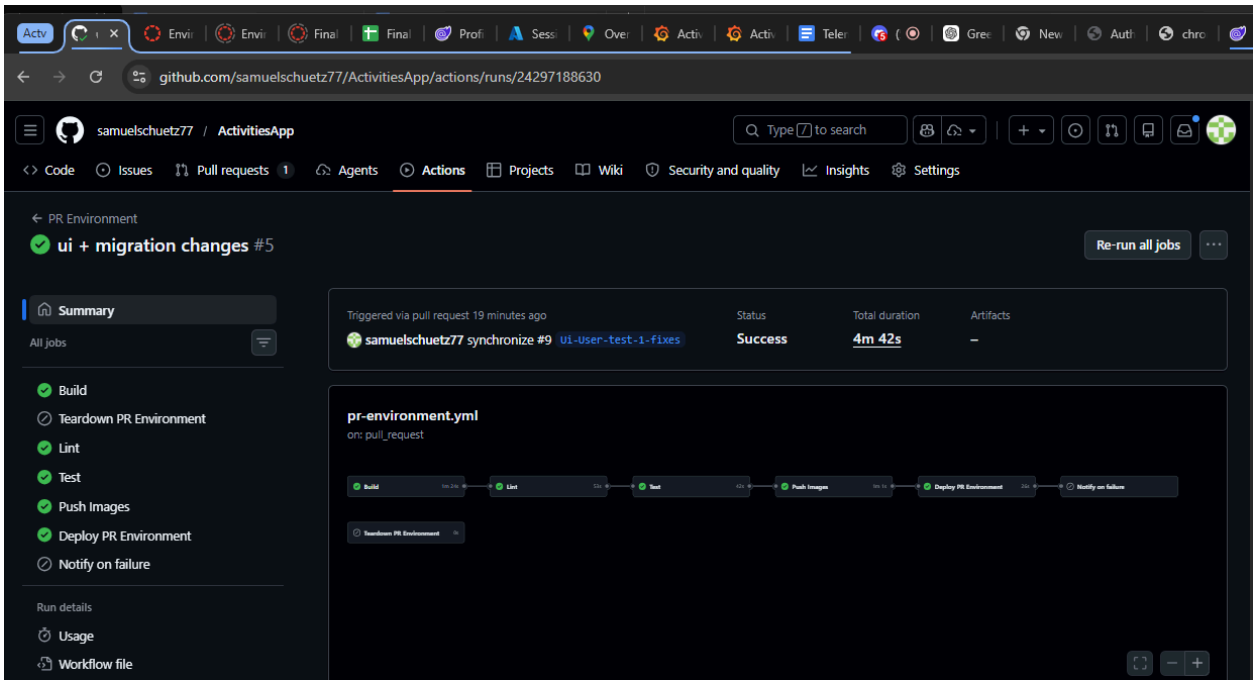
```
      SHA_TAG: ${{ needs.push.outputs.sha_tag }}
      NAMESPACE: pr-${{ github.event.pull_request.number }}
      PR_HOST: pr-${{ github.event.pull_request.number
}}.activor.duckdns.org
    steps:
      - uses: actions/checkout@v4

      - name: Set KUBECONFIG
        run: echo "KUBECONFIG=$HOME/.kube/config" >> "$GITHUB_ENV"

      - name: Verify cluster access
        run: kubectl get nodes

      - name: Create namespace
        run: |
          kubectl create namespace ${{ env.NAMESPACE }} --dry-run=client
-o yaml | kubectl apply -f -

      - name: Create secrets
        run: |
          kubectl create secret generic app-secrets \
            --namespace=${{ env.NAMESPACE }} \
            --from-literal=POSTGRES_PASSWORD='${{
secrets.POSTGRES_PASSWORD }}' \
            --from-literal=COSMOS_ENDPOINT='${{ secrets.COSMOS_ENDPOINT
}}' \
            --from-literal=COSMOS_KEY='${{ secrets.COSMOS_KEY }}' \
            --from-literal=GOOGLE_MAPS_API_KEY='${{
secrets.GOOGLE_MAPS_API_KEY }}' \
            --from-literal=AZURE_CLIENT_SECRET='${{
secrets.AZURE_CLIENT_SECRET }}' \
            --dry-run=client -o yaml | kubectl apply -f -

      - name: Deploy PostgreSQL
        run: |
          sed 's/namespace: activitiesapp/namespace: ${{ env.NAMESPACE
}}/g' deploy/k8s/data/postgres-pvc.yaml | kubectl apply -f -
          sed 's/namespace: activitiesapp/namespace: ${{ env.NAMESPACE
}}/g' deploy/k8s/data/postgres-dep.yaml | kubectl apply -f -
```

```
          sed 's/namespace: activitiesapp/namespace: ${{ env.NAMESPACE
}}/g' deploy/k8s/data/postgres-svc.yaml | kubectl apply -f -

      - name: Deploy API
        run: |
          sed 's/namespace: activitiesapp/namespace: ${{ env.NAMESPACE
}}/g' deploy/k8s/app/api-dep.yaml | kubectl apply -f -
          sed 's/namespace: activitiesapp/namespace: ${{ env.NAMESPACE
}}/g' deploy/k8s/app/api-svc.yaml | kubectl apply -f -

      - name: Deploy Web
        run: |
          sed 's/namespace: activitiesapp/namespace: ${{ env.NAMESPACE
}}/g' deploy/k8s/app/web-dep.yaml | kubectl apply -f -
          sed 's/namespace: activitiesapp/namespace: ${{ env.NAMESPACE
}}/g' deploy/k8s/app/web-svc.yaml | kubectl apply -f -

      - name: Create PR ingress
        run: |
          cat <<EOF | kubectl apply -f -
          apiVersion: networking.k8s.io/v1
          kind: Ingress
          metadata:
            name: activitiesapp-web
            namespace: ${{ env.NAMESPACE }}
            annotations:
              nginx.ingress.kubernetes.io/proxy-http-version: "1.1"
              nginx.ingress.kubernetes.io/proxy-read-timeout: "3600"
              nginx.ingress.kubernetes.io/proxy-send-timeout: "3600"
              nginx.ingress.kubernetes.io/ssl-redirect: "false"
              nginx.ingress.kubernetes.io/force-ssl-redirect: "false"
          spec:
            ingressClassName: nginx
            rules:
              - host: ${{ env.PR_HOST }}
                http:
                  paths:
                    - path: /
                      pathType: Prefix
                      backend:
```

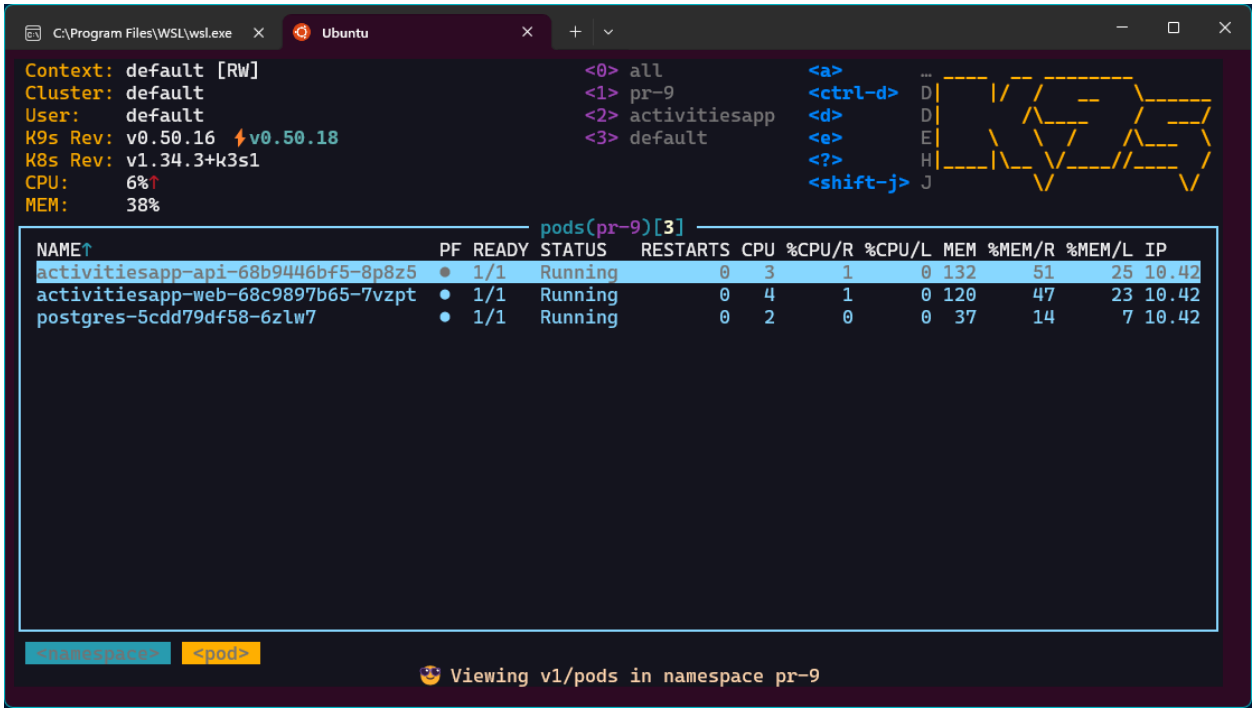
```
                        service:
                          name: activitiesapp-web
                          port:
                            number: 80
          EOF

      - name: Set deployment images
        run: |
          kubectl set image deployment/activitiesapp-api \
            api=${{ env.API_IMAGE }}:pr-${{ env.PR_NUMBER }}-${{
env.SHA_TAG }} \
            --namespace=${{ env.NAMESPACE }}
          kubectl set image deployment/activitiesapp-web \
            web=${{ env.WEB_IMAGE }}:pr-${{ env.PR_NUMBER }}-${{
env.SHA_TAG }} \
            --namespace=${{ env.NAMESPACE }}

      - name: Set APP_VERSION
        run: |
          kubectl set env deployment/activitiesapp-api APP_VERSION=pr-${{
env.PR_NUMBER }}-${{ env.SHA_TAG }} --namespace=${{ env.NAMESPACE }}
          kubectl set env deployment/activitiesapp-web APP_VERSION=pr-${{
env.PR_NUMBER }}-${{ env.SHA_TAG }} --namespace=${{ env.NAMESPACE }}

      - name: Wait for rollouts
        run: |
          kubectl rollout status deployment/activitiesapp-api
--namespace=${{ env.NAMESPACE }} --timeout=120s
          kubectl rollout status deployment/activitiesapp-web
--namespace=${{ env.NAMESPACE }} --timeout=120s

      - name: Post PR comment with URL
        run: |
          cat > /tmp/pr-comment.json <<'ENDJSON'
          {
            "body": "**PR Environment deployed!**\n\n**URL:** http://${{
env.PR_HOST }}\n\n**Namespace:** `${{ env.NAMESPACE }}`\n**API Image:**
`${{ env.API_IMAGE }}:pr-${{ env.PR_NUMBER }}-${{ env.SHA_TAG }}`\n**Web
Image:** `${{ env.WEB_IMAGE }}:pr-${{ env.PR_NUMBER }}-${{ env.SHA_TAG
}}`"
```

```
          }
          ENDJSON
          curl -s -X POST \
            -H "Authorization: token ${{ secrets.GITHUB_TOKEN }}" \
            -H "Accept: application/vnd.github.v3+json" \
            -H "Content-Type: application/json" \
            "https://api.github.com/repos/${{ github.repository
}}/issues/${{ env.PR_NUMBER }}/comments" \
            -d @/tmp/pr-comment.json

      - name: Post-deploy status
        if: always()
        run: |
          echo "=== Pods ==="
          kubectl get pods --namespace=${{ env.NAMESPACE }}
          echo ""
          echo "=== Services ==="
          kubectl get svc --namespace=${{ env.NAMESPACE }}
          echo ""
          echo "=== Ingress ==="
          kubectl get ingress --namespace=${{ env.NAMESPACE }}

  # ── Teardown on PR close ──

  teardown:
    name: Teardown PR Environment
    if: github.event.action == 'closed'
    runs-on: self-hosted
    env:
      NAMESPACE: pr-${{ github.event.pull_request.number }}
    steps:
      - uses: actions/checkout@v4

      - name: Set KUBECONFIG
        run: echo "KUBECONFIG=$HOME/.kube/config" >> "$GITHUB_ENV"

      - name: Delete PR namespace
        run: |
          kubectl delete namespace ${{ env.NAMESPACE }}
--ignore-not-found=true
```

```
      - name: Post teardown comment
        run: |
          curl -s -X POST \
            -H "Authorization: token ${{ secrets.GITHUB_TOKEN }}" \
            -H "Accept: application/vnd.github.v3+json" \
            -H "Content-Type: application/json" \
            "https://api.github.com/repos/${{ github.repository
}}/issues/${{ env.PR_NUMBER }}/comments" \
            -d '{"body":"**PR Environment torn down.** Namespace `${{
env.NAMESPACE }}` and all resources have been deleted."}'
```

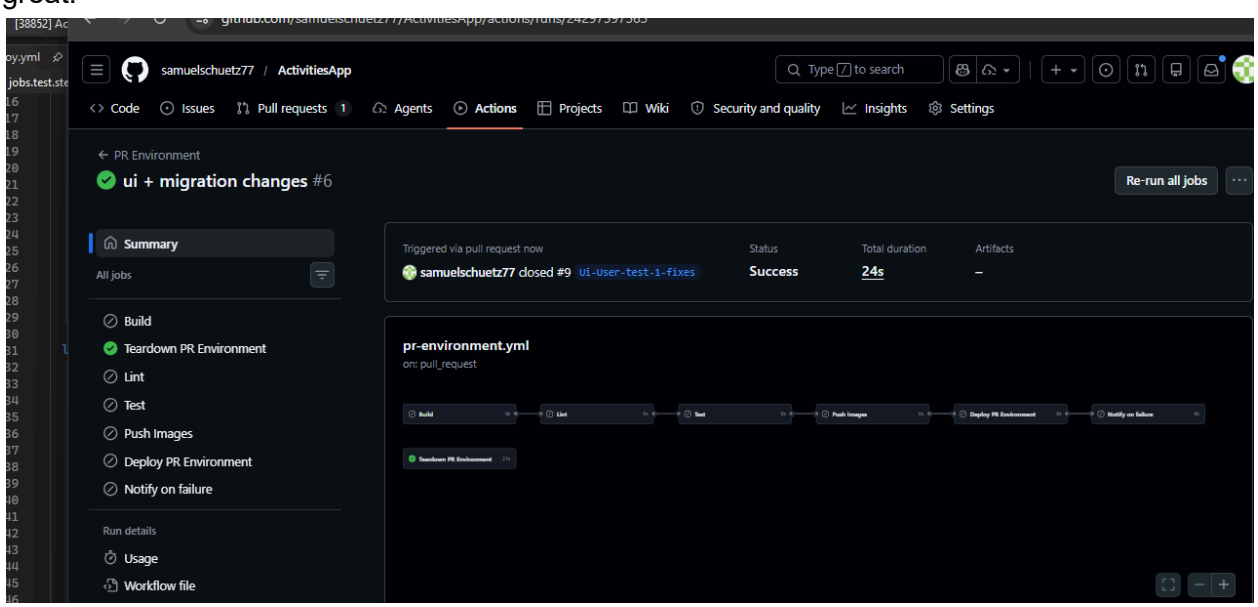


k9s showing the resources in kubernetes for this new environment:



Description:

Pipeline runs good but looks kinda bloated. We couldn't use gh because we don't have sudo on the runners, so it looks bad because it is using raw api calls instead. Tear down also worked great:



It deleted our namespaces and the url 404e

****If it doesn’t let you view the page try these instructions (for chrome):

Paste `chrome://net-internals/#hsts` into your address bar.

1. Scroll down to the **"Delete domain security policies"** section.
2. Type `activor.duckdns.org` into the domain box and click **Delete**.
3. Try visiting http://pr-9.activor.duckdns.org again.